RevoBank Strategy: Enhancing Credit Card Usage Among Existing Customers

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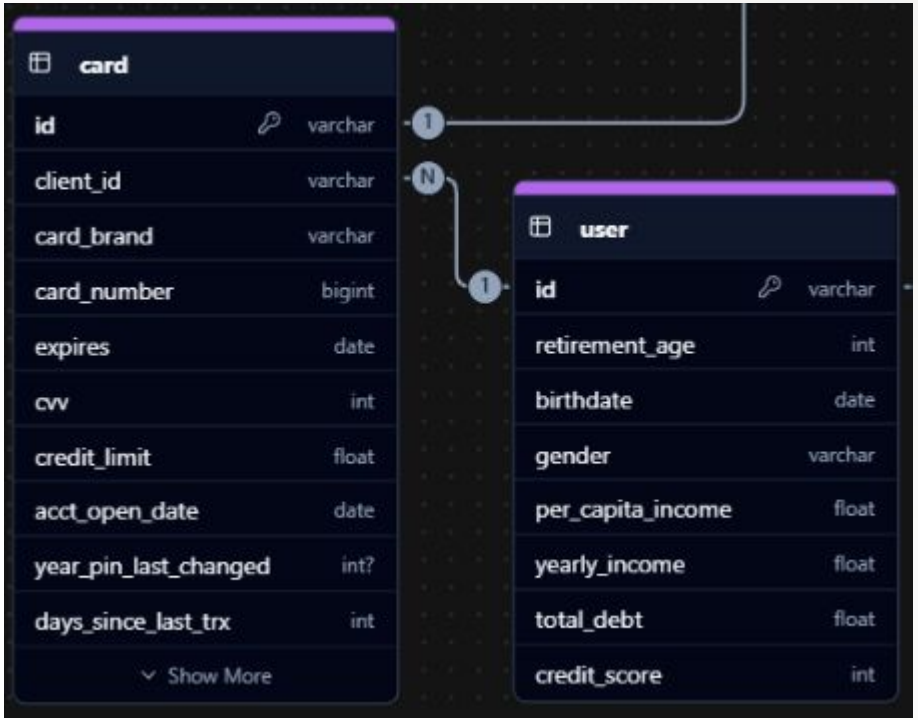
01

Business Overview

Business Overview

RevoBank is a leading bank in Indonesia offering various financial services, including credit cards. With a diverse customer base, RevoBank's goal is to maximize engagement and promote the use of credit cards among existing customers. Currently, many customers have credit cards that they do not utilize frequently. To drive growth and profitability, the bank aims to boost the frequency of credit card usage through strategic marketing initiatives and personalized customer engagement.

ERD : Data Dictionary



Disclaimer:

- Revo Bank is a fictional entity and the result presented here are purely for educational purposes.
- The queries and insight generated in this assignment should be personalized to my own analysis results.

Project Statement

"How can RevoBank increase its customers' credit card usage by 10% in 6 months through targeted marketing strategies and improved user engagement?"

Objective

AIM	To analyze sales performance over the past 6 months and create detailed user personas based on existing customers' demographics and Debt-to-Income ratio (DTI).
THE GOAL	To increase the frequency of RevoBank credit card usage among existing customers by implementing targeted strategies.

Methodology

Change Data Type	Change ID to string, numerical values to integers, and monetary amounts to floats.
Handle Typo	Clean typos by reviewing each account value and using the replace function.
Delete Duplicate Data	After identifying account values, search for duplicated data and remove irrelevant entries.
Fill and Remove Missing Value	After identifying null values in the data, either remove irrelevant columns or fill in the missing data.
Remove expired and 0 value cards	Exclude rows with cards that expired before May 31, 2025, and have a credit limit of 0.
Add columns	Create new columns for age, retirement age, and DTI.
Add Age Segmentation	Make Age Segmentation bins
Add Cluster	Make Cluster using K-means

02

Cleaning Data

Data Cleaning Step by Step

Import library panda and csv file in python

Make df_copy and rename dataframe copy

Changes data type

Using value count in each column to search for typo and null data

Correct typo with replace in card_brand 'Visa ' to Visa and 'Jcb' to 'JCB'

drop duplicated data in column ID

Fill credit_limit, count_nonfraud_trx_L6M, amnt_nonfraudtrx_L6M, count_fraudtrx_L6M, and amnt_fraudtrx_L6M with 0

Filter data using credit_limit and expires

Python : Data Analysis

Card Dataset

Python : Data Analysis

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5599 entries, 0 to 5598
Data columns (total 14 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   id                      5599 non-null   int64
 1   client_id               5599 non-null   int64
 2   card_brand              5599 non-null   object
 3   card_number             5599 non-null   int64
 4   expires                 5599 non-null   object
 5   cvv                     5599 non-null   int64
 6   credit_limit            5587 non-null   object
 7   acct_open_date          5599 non-null   object
 8   year_pin_last_changed   5599 non-null   int64
 9   days_since_last_trx     5599 non-null   int64
 10  count_nonfraud_trx_L6M  3707 non-null   float64
 11  amt_nonfraud_trx_L6M    3707 non-null   object
 12  count_fraud_trx_L6M     547 non-null    float64
 13  amt_fraud_trx_L6M       547 non-null    object
dtypes: float64(2), int64(6), object(6)
memory usage: 612.5+ KB
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 5518 entries, 0 to 5567
Data columns (total 14 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   id                      5518 non-null   object
 1   client_id               5518 non-null   object
 2   card_brand              5518 non-null   object
 3   card_number             5518 non-null   int64
 4   expires                 5518 non-null   datetime64[ns]
 5   cvv                     5518 non-null   int64
 6   credit_limit            5518 non-null   float64
 7   acct_open_date          5518 non-null   datetime64[ns]
 8   year_pin_last_changed   5518 non-null   int64
 9   days_since_last_trx     5518 non-null   int64
 10  count_nonfraud_trx_L6M  5518 non-null   float64
 11  amt_nonfraud_trx_L6M    5518 non-null   float64
 12  count_fraud_trx_L6M     5518 non-null   float64
 13  amt_fraud_trx_L6M       5518 non-null   float64
dtypes: datetime64[ns](2), float64(5), int64(4), object(3)
memory usage: 646.6+ KB
```

Data Cleaning Steps	Original Data	After Cleaning	Reason to do this data cleaning step
Convert to the correct data type	ID, client_id= integer Expires, acct_open_date, credit_limiit = object	ID, client_id= object Expires, acct_open_date = datetime credit_limiit = float	To help when analysis the dataset. I also change ID because it act as an identifier and not intended for aggregation
Remove Irrelevant Data	credit_limit, amt_nonfraud_trx_L6M, amt_fraud_trx_L6M = Rp in front of number There's typo in card brand, visa , and Jcb. Card brand has 6 categories	credit_limit, amt_nonfraud_trx_L6M, amt_fraud_trx_L6M = there's no Rp in front of number Card brand only has 4 categories	To help when analysis the dataset
Treat Missing Value	credit_limit = 12 count_nonfraud_trx_L6M=1884 amt_nonfraud_trx_L6M=1884 count_fraud_trx_L6M=5021 amt_fraud_trx_L6M=5021	credit_limit=0 count_nonfraud_trx_L6M=0 amt_nonfraud_trx_L6M=0 count_fraud_trx_L6M=0 amt_fraud_trx_L6M=0	I fill the blank with 0 because I need that column for analysis. Even though 2 column has more than 89.67% null.
Treat Duplicated	Index = 5599 row ID = 62 rows duplicated	Index = 5568 row	ID in this case should not had duplicate, so I drop and keep the first one.
Remove all expired date	Index = 5568 row	Index = 5518 row	To ensure data is up to date

User Dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   id                 2000 non-null   int64
 1   retirement_age     2000 non-null   int64
 2   birthdate          2000 non-null   object
 3   gender             2000 non-null   object
 4   per_capita_income  2000 non-null   object
 5   yearly_income      2000 non-null   object
 6   total_debt         2000 non-null   object
 7   credit_score       2000 non-null   int64
dtypes: int64(3), object(5)
memory usage: 125.1+ KB
```

```
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   id                 2000 non-null   object
 1   retirement_age     2000 non-null   int64
 2   birthdate          2000 non-null   datetime64[ns]
 3   gender             2000 non-null   object
 4   per_capita_income  2000 non-null   float64
 5   yearly_income      2000 non-null   float64
 6   total_debt         2000 non-null   float64
 7   credit_score       2000 non-null   int64
 8   age                2000 non-null   int64
 9   retired_flag       2000 non-null   bool
 10  DTI                2000 non-null   float64
```

Data Cleaning Steps	Original Data	After Cleaning	Reason to do this data cleaning step
Convert to the correct data type	ID = integer Birthday = object Per_capita_income, yearly_income, total debt = object	ID = object Birthday = datetime Per_capita_income, yearly_income, total debt = float	To help when analysis the dataset
Remove Irrelevant Data	Per_capita_income, yearly_income, total debt= Rp in front number	Per_capita_income, yearly_income, total debt= only number	To help when analysis the dataset
Add Column Age	none	There's column age based on 31 May 2025 - birthdate	Searching for demography customer
Add Column Retired Flag	none	There's column retired flag based on retired age and age	Searching for demography customer
Add Column DTI	none	There's column DTI using (total debt/per_capita_income) *100	To know about relation customer income amount with their debt

03

Analysis Data

Analysis Data Step by Step

Merge 2 dataset: card and user

There's 2 ID: id_x and id_y. Delete id_y, rename id_x to id

Filter transaction <=182 days (6 months)

Search MDR Profit using Net Profit=(Total Sales×MDR Rate)−Total Fraud Amount

Search Fraud Rate using fraud_rate = (total_fraud / (total_fraud + total_nonfraud)) * 100

Search Transaction Behaviour by Aggregated count and amount non fraud with average

Search for DTI rate =(Total Debt/Yearly Income)x100

Drop column per_capita_income, cvv, acct_open_date, and card_number

Calculate avg. DTI based on Credit Score

Search relationship on Age and Credit Limit

Make K-means Clustering

Merged Dataset Card and User

0	id_x	5528 non-null	object
1	client_id	5528 non-null	object
2	card_brand	5528 non-null	object
3	card_number	5528 non-null	int64
4	expires	5528 non-null	datetime64[ns]
5	cvv	5528 non-null	int64
6	credit_limit	5528 non-null	float64
7	acct_open_date	5528 non-null	datetime64[ns]
8	year_pin_last_changed	5528 non-null	int64
9	days_since_last_trx	5528 non-null	int64
10	count_nonfraud_trx_L6M	5528 non-null	float64
11	amt_nonfraud_trx_L6M	5528 non-null	float64
12	count_fraud_trx_L6M	5528 non-null	float64
13	amt_fraud_trx_L6M	5528 non-null	float64
14	id_y	5528 non-null	object
15	retirement_age	5528 non-null	int64
16	birthdate	5528 non-null	datetime64[ns]
17	gender	5528 non-null	object
18	per_capita_income	5528 non-null	float64
19	yearly_income	5528 non-null	float64
20	total_debt	5528 non-null	float64
21	credit_score	5528 non-null	int64
22	age	5528 non-null	int64
23	retired_flag	5528 non-null	bool
24	DTI	5528 non-null	float64

#	Column	Non-Null Count	Dtype
0	id	3679 non-null	object
1	client_id	3679 non-null	object
2	card_brand	3679 non-null	object
3	expires	3679 non-null	datetime64[ns]
4	credit_limit	3679 non-null	float64
5	year_pin_last_changed	3679 non-null	int64
6	days_since_last_trx	3679 non-null	int64
7	count_nonfraud_trx_L6M	3679 non-null	float64
8	amt_nonfraud_trx_L6M	3679 non-null	float64
9	count_fraud_trx_L6M	3679 non-null	float64
10	amt_fraud_trx_L6M	3679 non-null	float64
11	retirement_age	3679 non-null	int64
12	birthdate	3679 non-null	datetime64[ns]
13	gender	3679 non-null	object
14	yearly_income	3679 non-null	float64
15	total_debt	3679 non-null	float64
16	credit_score	3679 non-null	int64
17	age	3679 non-null	int64
18	retired_flag	3679 non-null	bool
19	DTI	3679 non-null	float64
20	DTI_ratio	3679 non-null	float64
21	score_category	3679 non-null	category
22	age_group	2962 non-null	category
23	age_segment	2962 non-null	category
24	cluster	3679 non-null	int32

Milestone 2

Revenue & Profitability (MDR Profit)

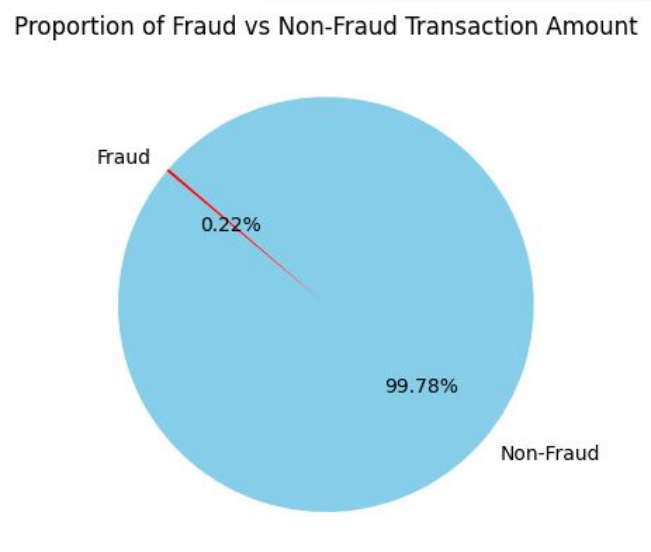
Rp 5.8B MDR profit is GOOD

it ‘s means promos worked

This came from 46-60 year olds (169B spending) - RevoBank VIP cash cows

RevoBank’s profitability is volume-driven. As MDR revenue is earned as a percentage of transaction value, overall profit is highly dependent on sustained card usage among high-income customers, who contribute a disproportionate share of total spending.

The Whale-Hunting Risk: Risk Management (Fraud & DTI)



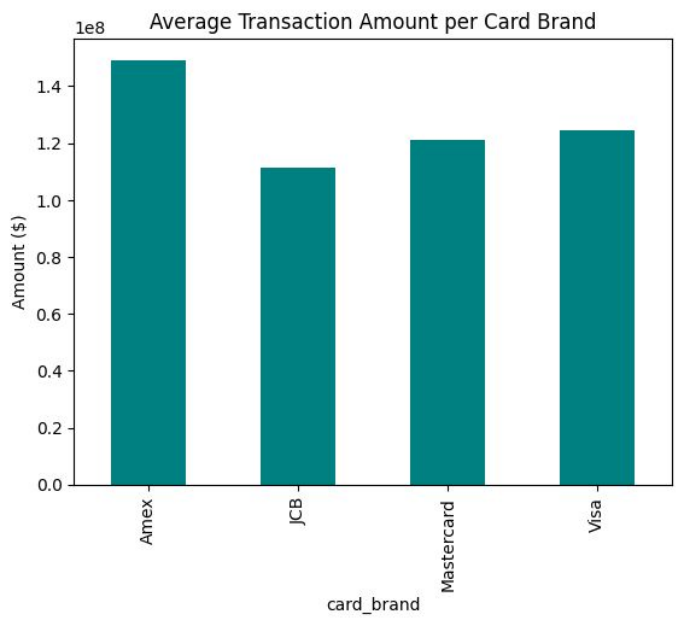
🚨Watch TOP 10 Fraud Cases (Rp26M total)

0.22% fraud rate = EXCELLENT (industry avg 1%) but high outliers (26.9M). DTI outliers at 80%.

Recommendations:

- **Implement enhanced security controls** (e.g., manual 2FA or phone verification) for transactions exceeding Rp 15M to mitigate high-value fraud risk.
- **Offer debt consolidation loans** to customers with DTI above 80%, converting high-interest revolving card debt into structured, lower-risk installment loans.
- **Strengthen monitoring of high-risk customers** through advanced fraud detection and targeted client education on security best practices.

Brand Positioning: Volume vs. Value
Card Brand Strategy (Mastercard vs. Amex)



Mastercard has a broader user base; Amex users have higher satisfaction and usage frequency

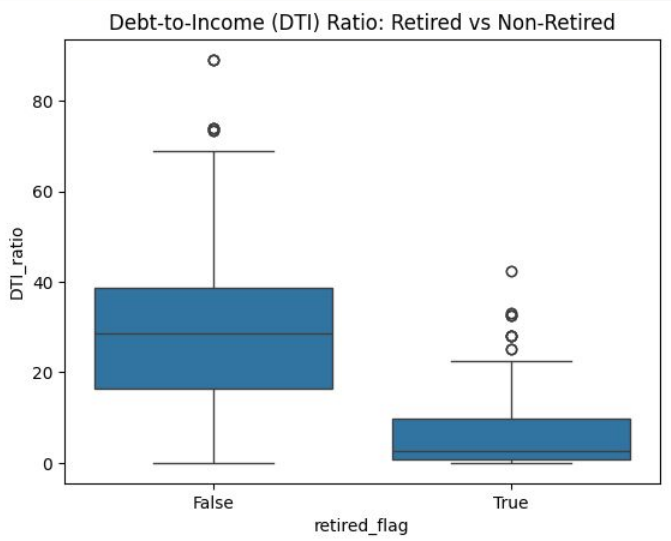
American Express cards have fewer users but significantly higher spending per customer, indicating a premium customer segment. Given that Amex typically carries higher MDR fees, each Amex transaction is more profitable per swipe compared to Mastercard.

Recommendation:

- **Implement targeted marketing and retention strategies** for Amex cardholders,
- **Emphasizing premium benefits** to increase usage frequency and protect high-margin revenue.

DTI Retired vs Working Ratio

Working people: Risky (28% DTI), Retired: Safe (low DTI)



🚨 Watch 80%+ DTI workers - they can't handle emergencies

Non-Retired Clients :

- Result: Higher median DTI indicates greater credit exposure and financial pressure.
- **Recommendation**: **Provide proactive financial guidance and debt management tools** to reduce long-term credit risk.

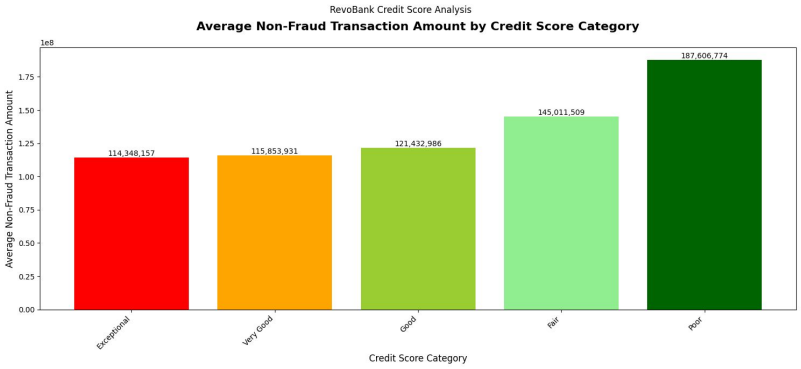
Retired Clients

- Result: Lower DTI reflects stable financial behavior and controlled leverage.
- **Recommendation: Offer conservative, income-oriented products** such as investment or savings accounts aligned with retirement needs.

High-Risk Outliers

- Result: A subset of customers exhibits DTI above 80%, signaling elevated default risk.
- **Recommendation**: **Closely monitor** these accounts and **actively recommend debt restructuring or consolidation solutions**.

DTI Credit Score



Poor Category are high spenders (Rp 187M)

Poor Category:

Result: Poor-credit customers generate high transaction volume but exhibit elevated financial risk, indicating reliance on credit beyond sustainable levels.

Recommendation: Immediately **tighten daily spending limits and introduce financial education** initiatives to prevent further overextension.

Fair Category:

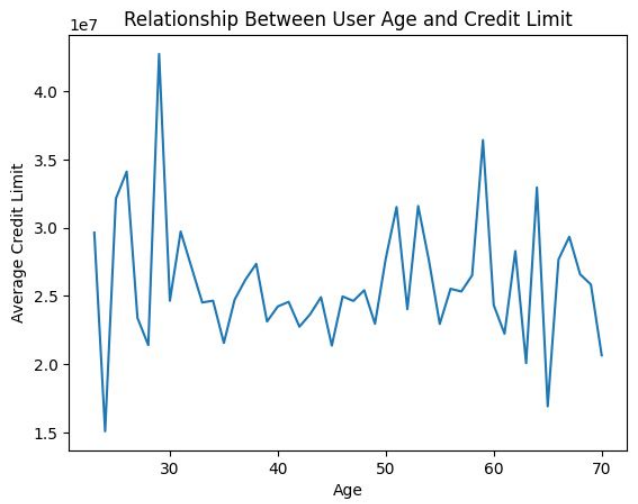
Result: Fair-credit customers display high DTI, signaling ongoing financial stress.

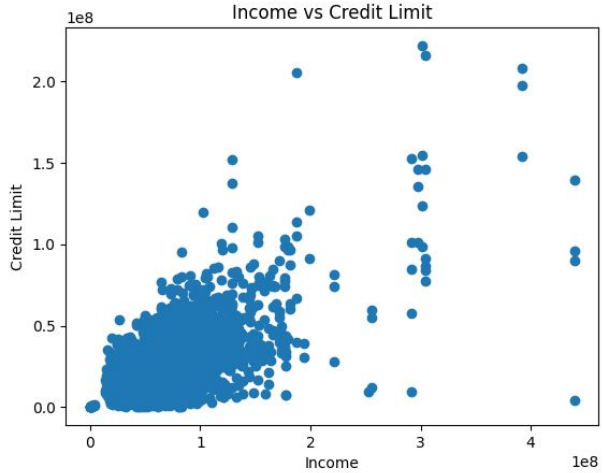
Recommendation: **Offer personalized repayment plans or structured loan products** to stabilize their financial position.

Good & Above:

Result: Customers with Good or higher credit scores demonstrate stable spending behavior and low DTI, indicating strong repayment capacity.

Recommendation: Encourage more frequent card usage through tactical incentives such as weekend multipliers or small-ticket rewards, while **reinforcing loyalty with exclusive benefits**.





Age and Credit Limit

The observed peaks in average credit limits among younger customers suggest a forward-looking lending strategy, reflecting the bank's confidence in their future earning potential.

A positive correlation between income and credit limit is observed, indicating that higher-income customers are perceived as lower risk and are therefore granted higher credit limits. This alignment reflects sound risk-based lending practices.

Age Segment Strategy

age_segment	credit_limit	amt_nonfraud_trx_L6M	DTI_ratio
Early Career	27593260.00	17160698800.00	27.24
Family & Establishment	24492634.90	130717808800.00	27.95
Peak Income	26463847.81	169477103100.00	28.18
Pre-Retirement	25652727.27	70263691600.00	21.79

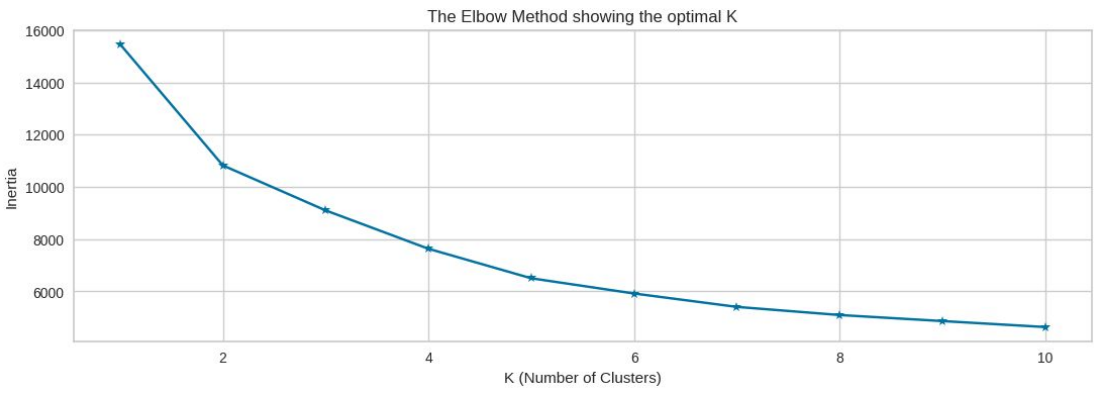
Age Segment	Result	Recommendation
Early Career (21-30) (Anomaly/Lazy Capital | 57 Clients)	High limits, low use	**Decrease credit limits** by Rp 2–3M per customer, or approximately 20%, to reduce unfunded exposure and mitigate downside credit risk.
Family & Establishment (31-45) (Stretched Group/The Bottleneck | 360 Clients)	Low limits, high spend	**Implement a targeted Credit Limit Increase** (CLI) campaign based on life-stage and income progression to capture additional transaction volume and unlock incremental MDR revenue.
Peak Income (46-60) (Cash Cows | 424 Clients)	Highest spending (Rp 169.5 Billion).	**Treat top-spending** customers as **VIPs through exclusive promotions** such as travel rewards and concierge services, ensuring retention and protecting this high-value revenue segment from competitor attrition.
Pre-retirement (61-71) (Safe Haven | 186 Clients)	Lowest DTI, low credit risk	**Offer low-interest balance transfer options** to improve financial efficiency while deepening customer engagement.

Milestone 3

K-Means Cluster

```
KMeans
KMeans(n_clusters=3, random_state=42)
```

RFM captures recency and frequency but ignores credit risk and capacity. K-Means enables simultaneous segmentation based on transaction behavior, leverage, and credit exposure.

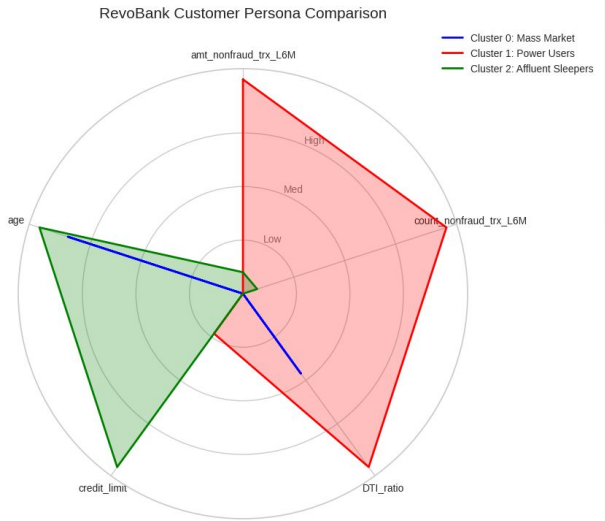


For k=2, the average silhouette score is 0.43073546851146527
For k=3, the average silhouette score is 0.2940526873872515
For k=4, the average silhouette score is 0.2435310936159183
For k=5, the average silhouette score is 0.24820580330480616
For k=6, the average silhouette score is 0.244891895697072

Based on Elbow Method and Silhouette Score, I choose to make 3 Cluster. I used Robust Scaler because there's many Outlier in the Data.

Phyton

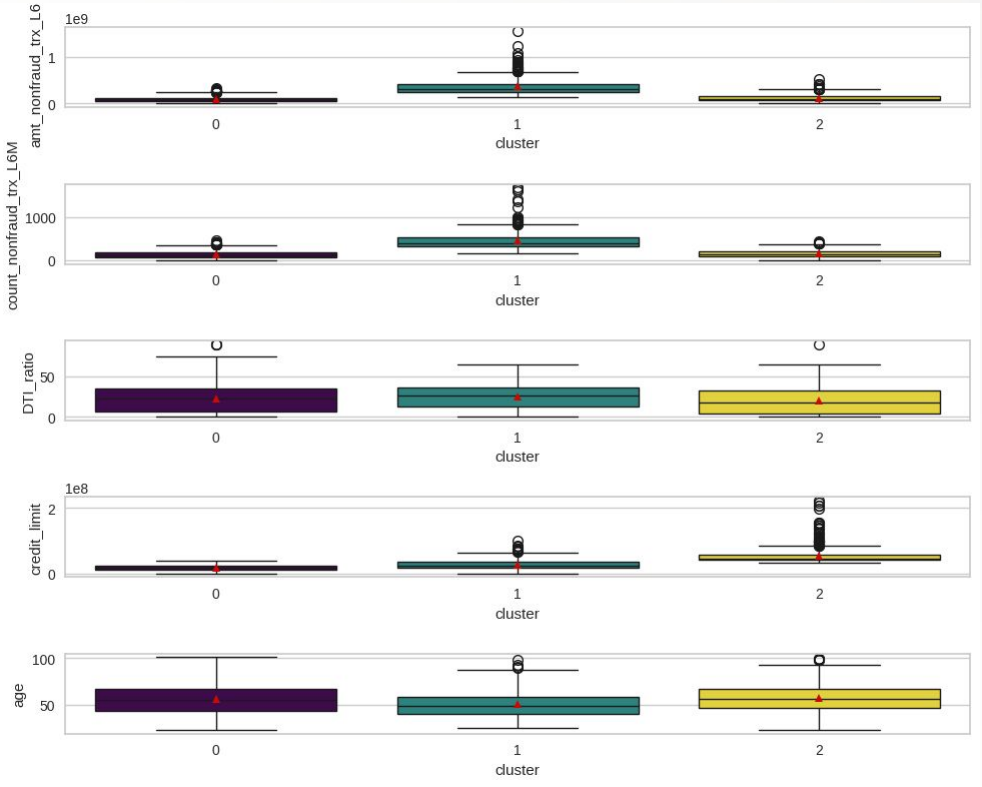
User Persona



Persona	Key Characteristic	Suggest Action
Cluster 0: Stable Mass Market	Low Limit / Stable	Monitor: Keep operational costs low.
Cluster 1: Power Users	High Activity / High Risk	Retain: Protect this revenue with loyalty rewards.
Cluster 2: Affluent Sleepers	High Capacity / Low Usage	Activate: Use premium perks to "wake up" their spending.

Phyton

Key Finding



- **Significant Separation**: The boxes for Spending and Frequency do not overlap between Cluster 1 and Cluster 0, proving the **segments are mathematically distinct**.
- **Low-Variance Safety**: **Cluster 2** (Seniors/Sleepers) shows very tight distributions in DTI, confirming they **are our safest lending target**.
- **Profitability Outliers**: **Cluster 1** contains extreme "Whale" spenders (dots at >Rp1.0B) that **drive a disproportionate share of MDR profit**.

Phyton

Cluster	Insight	Interpretation	Recommendation
0 (Stable) (986 Clients | Profit Rp2,46B)	Spend: Low to Medium, Frequency: Low to Medium, DTI Ratio: Medium, Credit Limit: Low, Age: Medium to High	It has the most customer count that older but contributes stable even though it had limited revenue, driven by conservative usage behavior.	• Fee-based products (insurance, bill pay) • No aggressive credit expansion • Focus on digital self-service and low-cost maintenance.
1 (Active Spenders) (404 Clients | Profit Rp2,48B)	Spend: High, Frequency: High, DTI Ratio: Medium, Credit Limit: Medium, Age: Low to Medium	The most active and profitable segment per customer	• Premium rewards optimization • Early-warning DTI monitoring (soft controls) • Offer "Everyday Value" rewards (groceries/fuel)
2 (Unused Credit) (350 Clients | Profit Rp873M)	Spend: Low to Medium, Frequency: Medium, DTI Ratio: Medium, Credit Limit: High, Age: High	Wealthy individuals with the highest credit limits but had low than expected activity.	• Spend activation campaigns • Periodic credit line review • Use premium category-based incentives offers (travel, dining, healthcare, luxury perks)

04

Summary and Recommendation

Revenue is primarily driven by transaction behavior rather than credit exposure, Cluster 1 serves as the primary revenue catalyst, while Cluster 2 represents underutilized capital with high growth potential, and Cluster 0 provides stable but limited contribution, driven by conservative spending behavior.

Focus in 2 Cluster :

- **Cluster 1 – Active Spenders :** Client spend 3.2× more than Cluster 2 on average and generate the highest MDR contribution.
- **Cluster 2 – Unused Credit :** Holds the highest average credit limit (Rp 53.9M) and the lowest average DTI (~20%), indicating strong credit quality.

Priority for Action:

1. Cluster 1 (Active Spenders) Family & Establishment (31–45) and Peak Income (46–60) Implement "Everyday Value" rewards
2. Cluster 2 (Unused Credit) Peak Income (46–60) and Pre-retirement (61–71) Launch a "Premium Lifestyle" campaign
3. Cluster 2 (for increase) and Cluster 1 (for monitoring) Family & Establishment (31–45) Deploy Life-Stage Credit Limit Increases

Thank You

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